

images from mathematical formulae can easily be generated with the help of a numerical evaluator of mathematical expressions in Clmg. There are many functions to generate 3D vector objects from dense image data.

3D object viewer. The framework is composed of a 3D object viewer. For 3D rendering, Clmg does not rely on external libraries.

Entire library under one header

If you are a professional C++ programmer, perhaps you would be surprised to notice that the entire library is written under a single header file. A bulky code might be time-consuming, but arranging these under different header files is not the best solution, as you might require all of these.

Of course, you can get the library precompiled. Yet again, this is not a good option because of the generic nature of the library.

Library is based on template data types. As in the case of a standard template library (STL), it is not possible to precompile library classes and functions of all possible data types.

Clmg library is based on template data types, where users have the freedom to opt for the type of image that is represented by their respective codes. There will be a lot of possible combinations of classes, functions, arguments and data types, and the size of the object binary file generated to cover all possible cases will be huge. Compiling on the fly with users' codes, in turn, introduces a lot of flexibility for the users while coding.



Fig. 2: Image restoration and inpainting using Tschumperlé-Deriche algorithm

Supported platforms

The software is highly portable. It has been successfully compiled under the following configurations:

- PC Linux 32-bit, with g++
- PC Windows 32-bit, with Visual C++ 6.0
- PC Windows 32-bit, with Visual C++ Express Edition
- Sun SPARC Solaris 32-bit, with g++
- Mac PPC with OS X and g++

Although external functionalities and libraries can be used to do more specific tasks, the minimal version of Clmg can be compiled only with standard C++ headers.

G'MIC: A full-featured image-processing framework

Despite its simplicity and usefulness, Clmg toolbox is not devoid of drawbacks. One main disadvantage is the prerequisite of knowing C++. Some users who would be interested in some functionalities of Clmg might not have sufficient knowledge of C++.

GREYC's Magic for Image Computing (G'MIC) is a full-featured image-processing framework based on Clmg library. It provides many user interfaces (UIs) convert, manipulate, filter or visualise generic image datasets in varying dimensions. We shall have a detailed discussion on this tool in a later issue of EFY Plus.

Interdependent classes. There are only four classes used in the library. Out of these, Clmg and ClmgList representing an image and a collection of image are the most important. Both these classes are interdependent. Even if we are defining these under different headers, we have to include both. Therefore in practice we do not gain anything in defining these under different header files.

Simplicity guaranteed. When the entire code is put under a single header, at first sight it may look like a total mess. But after thorough observation, we would realise that these are well-organised and structured.

From the user's perspective, installation is as simple as putting Clmg header in the source directory.

For the developer, maintenance of the code is pretty easy. If the developer is in search of a piece of code, he or she can easily spot it in the library.

Users' reviews

Reviews on websites like <http://sourceforge.net/> indicate that the users are quite happy with the tool. The generic nature of the tool to suit various applications, and the simplicity and flexibility it offers, attracts many users.

A user writes, "I have been using this library for developing simple demonstrative projects and for testing my ideas in a fast and efficient way. The library is simply amazing in what it offers to you as a single-header file with very little project headache."

Some users are concerned about documentation. "The tool is a remarkable application having C++ features. It is astonishing that it is usable with just the include file, without linking to external libraries. I do not know how this was achieved. It needs a lot of work on documentation, though. I spent weeks trying to figure out even very simple things, but there are people in the forum who can help in this regard," points out a user.

A cool image-processing framework

The framework is aptly named Clmg, an acronym for 'Cool Image.' The tool is not specific to any particular application, so anyone really interested in image processing can make use of it. The latest version of the software is provided in this month's EFY Plus DVD. Make sure to give it a try! ●

The author is an electronics enthusiast